L#03-04
Chapter-2

Mathematics In India
(IKS 4001)

Chapter 2: Vedic Geometry (Śulbasūtras)

2.1 Introduction
2.2 The Sulbasutra
2.3 The Theorem of Diagonal
2.4 Rectilinear Figures and their Transformation
2.5 Circle from Square: The Direct Construction
2.6 The Inverse Formula: Square from Circle

2.1 Introduction

- In this document, the author discusses the approach of examining Vedic geometry before exploring the geometrical knowledge of the Indus Valley civilization. The necessity for this approach stems from the availability of the Sulbasutras, which provide documented insights into Vedic geometric knowledge despite offering limited justification for their claims.
- In contrast, understanding Harappan geometry relies on indirect evidence, making it more conjectural. The author emphasizes that the Sulbasutras detail the architecture of Vedic rituals, specifying geometric forms and measurements used in building altars.
- These texts provide various geometric constructs, area transformations, and implicit theorems. The origins of the focus on geometry in rituals are uncertain, but early references are found in the Yajurveda.

- **Vedic Geometry developed for religious rituals**

➢ It was developed primarily for religious rituals, specifically the precise construction of fire altars (agnikundas) and sacrificial grounds (vedis). Priests believed different altar shapes offered different benefits, necessitating exact measurements.

- **Concerned mainly with construction problems**

➢ It is mainly concerned with practical construction problems rather than abstract theoretical proofs, which was a key distinction from later Euclidean geometry.

- **Foundations of Indian geometric thinking**

➢ The methods and rules described in texts like the Sulba Sutras (meaning "cord rules" or "science of measurement with a cord") are considered foundational to Indian geometric thinking. These texts contain early understandings of concepts equivalent to the Pythagorean theorem and methods for transforming shapes while preserving area (e.g., squaring the circle).

2.2 The Śulbasūtras

meaning of Śulba based on its role in ancient Indian geometry

- **Śulba = measuring cord**
  - The term comes from the Sanskrit root "śulb," meaning "to measure".
- **Geometry based on rope constructions**
  - Geometry based on rope constructions was documented in texts called the Śulbasūtras, which provided rules and procedures for priests to build sacrificial fire altars (agnikundas) with precise geometric shapes and areas.
- **Similar to modern straightedge methods**
  - A tightly stretched cord or rope could serve multiple functions, including acting as a straight edge, a compass (by drawing arcs around a point), and for creating proportional lengths, much like modern drafting and surveying tools.

Śulbasūtras

- The term 'sulba' refers to a cord or rope, while 's¯utra' describes a condensed writing style that omits non-essential words.

• 'Cord Sutras' pertain to geometry that can be constructed using inextensible cords, focusing on the fundamental elements of circles and straight lines.

• Two key principles in 'Sulbasutra' geometry are:

1. A circle is defined as the locus of points equidistant from a fixed center.
2. The shortest distance between two points is a straight line; this principle is rooted in the geometry of stretched cords.

- The concept of a geometrical point is implied through intersections of lines and arcs, not explicitly defined.

Short Answer Questions

- What does “Śulba” mean?
- State the Theorem of the Diagonal.
- Write the modern algebraic form.
- Why was this theorem important in Vedic ritual?

2.3 The Theorem of the Diagonal

Introduction

- The Theorem of the Diagonal appears in the Śulbasūtras, ancient Indian texts concerned with the geometric construction of Vedic ritual altars. The word 'Śulba' means rope, indicating that geometry was performed using cords for measurement and construction.
- This theorem is historically important because it represents the earliest known statement of what is now called the Pythagoras Theorem.

Learning Objectives

- Understand the historical background of the theorem.
- State the theorem in both traditional and modern forms.
- Explain the geometric meaning of the theorem.
- Apply the theorem to solve numerical problems.
- Recognize its importance in Vedic altar constructions.

WHO WAS BAUDHYANA?

- Boudhayan (800 BC - 740 BC) is said to be the original Mathematician behind the Pythagoras theorem. Pythagoras theorem was indeed known much before Pythagoras, and it was Indians who discovered it at least 1000 years before Pythagoras was born! The credit for authoring the earliest Sulba Sutras goes to him.

- The Sulbasutras is like a guide to the Vedas which formulate rules for constructing altars.
  In other words, they provide techniques to solve mathematical problems effortlessly.

WORK OF BAUDHYANA

- Boudhayan is credited with significant contributions towards the advancements in mathematics. The most prominent among them are as follows:
  - Circling a square.
  - Value of π
  - The method of finding the square root of 2.

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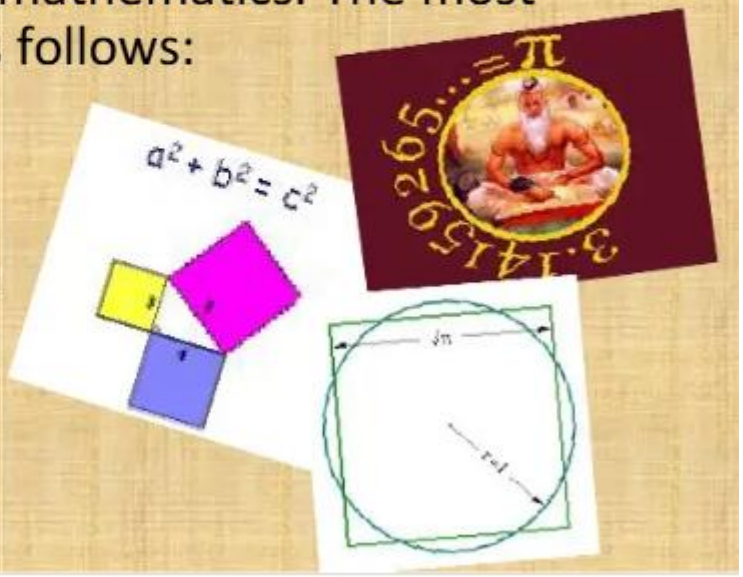
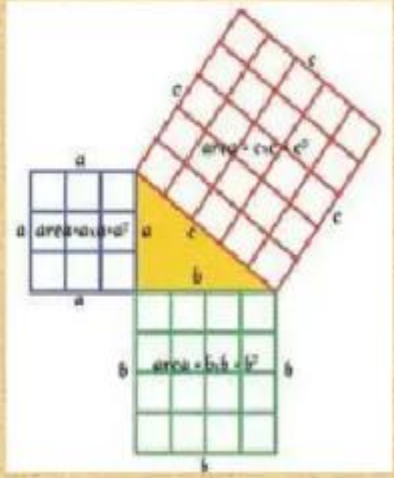


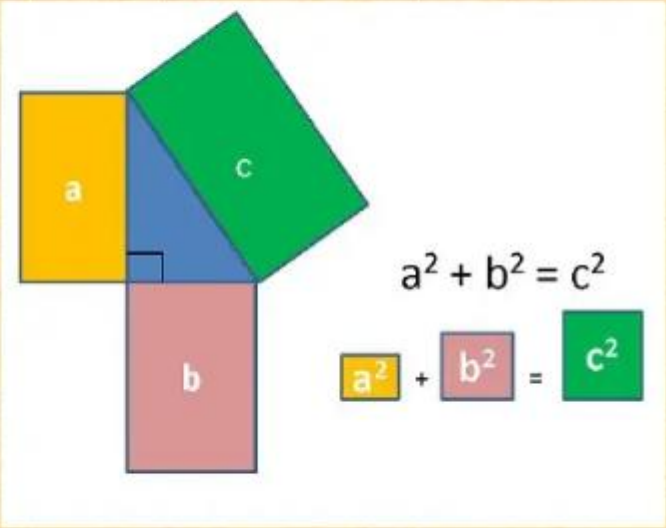
Boudhayan Theorem

- Theorem:- The diagonal of a rectangle produce by itself the same area as produced by its length and breadth
- The diagonal and sides referred to are those of a rectangle, and the areas are those of the squares having these line segments as their sides. Since the diagonal of a rectangle is the hypotenuse of the right triangle formed by two adjacent sides, the statement is seen to be equivalent to the Pythagoras theorem.



COMPARING IT WITH PYTHAGORAS THEOREM

- In any right-angled triangle, the area of the square whose side is the hypotenuse (the side opposite the right angle) is equal to the sum of the areas of the squares whose sides are the two legs (the two sides that meet at a right angle.
- c is the longest side of the triangle(this is called the hypotenuse) with a and b being the other two sides.



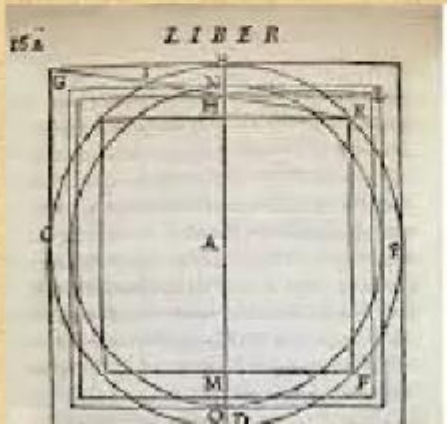
The rule stated by Boudhayan in Sanskrit is as follows-

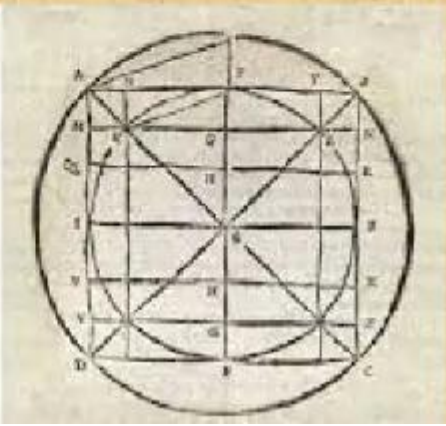
दीर्घचतुरश्रस्याक्ष्णया रज्जुः पार्श्वमानी तिर्यग् मानी च यत् पृथग् भूते कुरूतस्तदुभयं करोति ॥

Which means-

A rope stretched along the length of the diagonal produces an area which the vertical and horizontal sides make together In other words: $a^2 = b^2 + c^2$

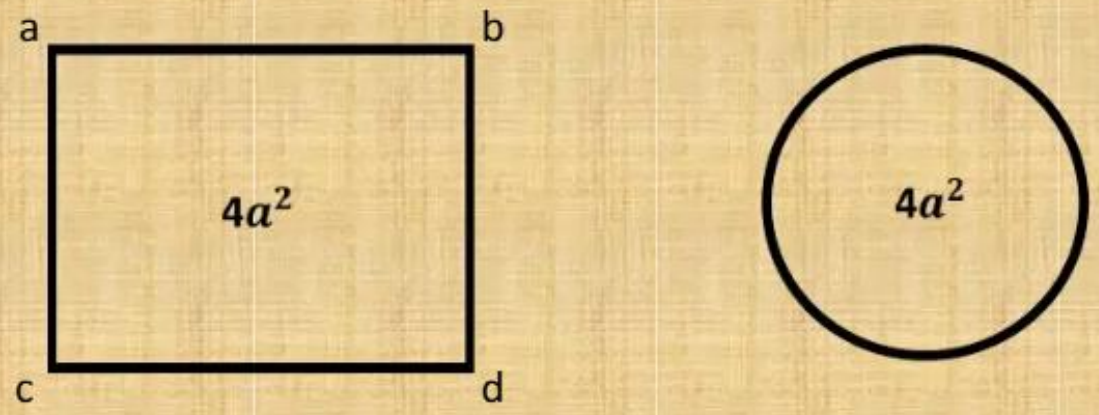
CONCEPT OF CIRCLING A SQUARE USING BOUDHYANA SULABHSUTRA





Circling of Square-

The meaning of circling a square is to draw a circle whose area is equal to the area of a Square.

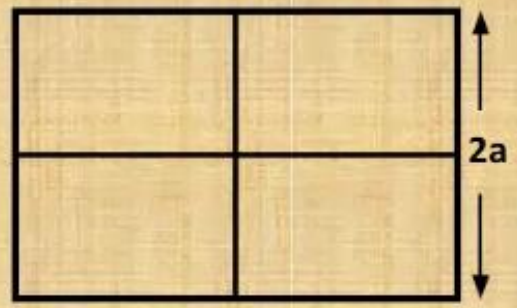


For circling of a square Boudhayan gave a formula using a sutra which says

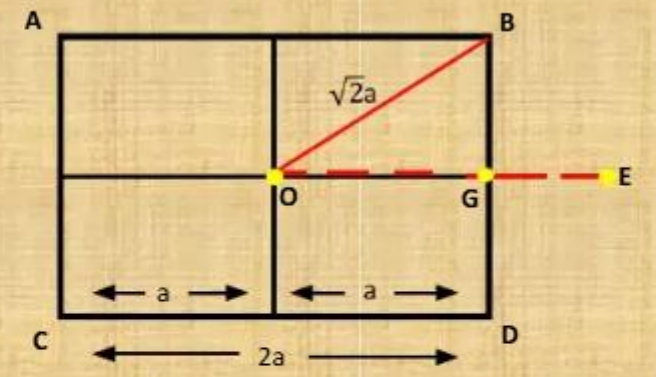
" If half the diagonal is stretched to the center towards the east and 1/3rd is added to the remainder, the circle drawn is equal to the area of the square"

Now, we will be discussing these steps of the sutra below-

- **STEP 1 –** Take a square of side 2a and divide it into four blocks

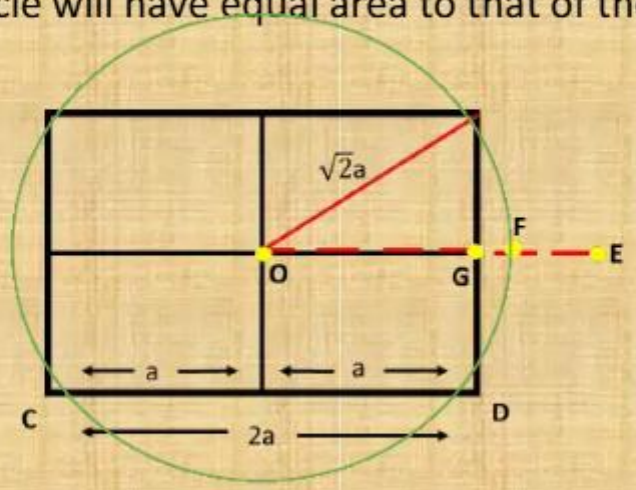


- **STEP 2** – Stretch half of the diagonal towards the center to the East



- OB = OE = Half of diagonal = $\sqrt{2}\,a$

❖ STEP 3 - Divide GE into 3 equal parts such that GF is 1/3rd of GE.
Finally take OF as a Radius and draw a circle.
This circle will have equal area to that of the Square.



$$GF = \frac{1}{3} GE$$

Proof of this Method ->

Area (A B C D) = $4a^2$

Radius of Circle = OF = OG +GF = a + $\frac{1}{3}GE$

= a + $\frac{1}{3}$(OE – OG)

= a + $\frac{1}{3}$($\sqrt{2}$a – a)

= $\frac{a}{3}$ $(2 + \sqrt{2})$

Radius = 1.1380716 . a

Value of $\sqrt{2}$ obtained by Boudhayan was **1.414215**

Area of circle = πr^2

$= \pi(1.1380716)^2\, a^2$

The value of π that Boudhayan got was **3.088**

$= \pi(1.1380716)^2\, a^2$

$= 3.088 \times (1.1380716)^2\, a^2$

$= 3.99\, a^2 \approx \mathbf{4\, a^2}$

Q-THE HYPOTENUSE OF A RIGHT-ANGLED TRIANGLE IS 16 UNITS AND ONE OF THE SIDES OF THE TRIANGLE IS 8 UNITS. FIND THE MEASURE OF THE THIRD SIDE USING THE PYTHAGORAS THEOREM FORMULA.

Answer:- Given: Hypotenuse = 16 units

Let us consider the given side of a triangle as the perpendicular height = 8 units

On substituting the given dimensions to the Pythagoras theorem formula

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Height}^2$$

$$16^2 = B^2 + 8^2$$

$$B^2 = 256 - 64$$

$$B = \sqrt{192} = 13.856 \text{ units}$$

Therefore, the measure of the third side of the triangle is 13.856 units.

Q-THE SIDES OF A TRIANGLE ARE 5, 12 & 13 UNITS. CHECK IF IT HAS A RIGHT ANGLE OR NOT.

Answer:- From Pythagoras Theorem, we have;

$$P^2 + B^2 = H^2$$

Let, Perpendicular (P) = 12 units

Base (B)= 5 units

Hypotenuse (H) = 13 units {since it is the longest side measure}

$$LHS = P^2 + B^2$$

$$\Rightarrow 12^2 + 5^2$$

$$\Rightarrow 144 + 25$$

$$\Rightarrow 169$$

$$RHS = H^2$$

$$\Rightarrow 13^2$$

$$\Rightarrow 169$$

$$\Rightarrow 169 = 169$$

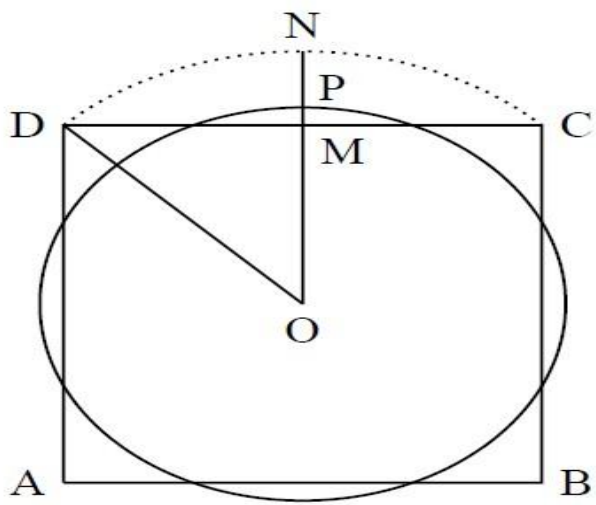
L.H.S. = R.H.S.

Therefore, the angle opposite to the 13 units side will be a right angle.

[OR]

Circle from Square: The Direct Construction

- Baudhayana gives a very interesting construction for a circle whose area is equal to that of a given square, of course approximately.



- We wish to transform the square *ABCD* into a circle with almost same area. Let *O* be the central point of the square (point of intersection of two diagonals).
- Let *M* be the midpoint of *CD*. Consider the line through *O* and *M*.
- Draw a circle with center *O* and radius *OD*. Suppose it meets the line through *O* and *M* at *N*. Divide *MN* at *P* such that $PM = \frac{1}{3}MN$.

- Now, construct a new circle with center *O* and radius *OP*. We see that the area of this circle is approximately equal in area to that of the square *ABCD*.
- Suppose each side of the square *ABCD* is $2a$. Then the area of this square is $4a^2$. By Bodhayana- Pythagoras theorem, the diagonal *BD* $=\sqrt{2}(2a)$. Hence $\boldsymbol{OD = \frac{1}{2}BD = \sqrt{2}a}$

Cont….

- In ΔOMD, $\angle M = 90^\circ$. Hence, by Bodhayana-Pythagoras theorem,
  $OM^2 = OD^2 - MD^2 = (\sqrt{2}a)^2 - a^2 = (2a)^2 - (a)^2 = a^2$
  Hence, **OM = a**
- Therefore, radius **r** = $OP = OM + MP = OM + \frac{1}{3}MN = OM + \frac{1}{3}(ON - OM)$

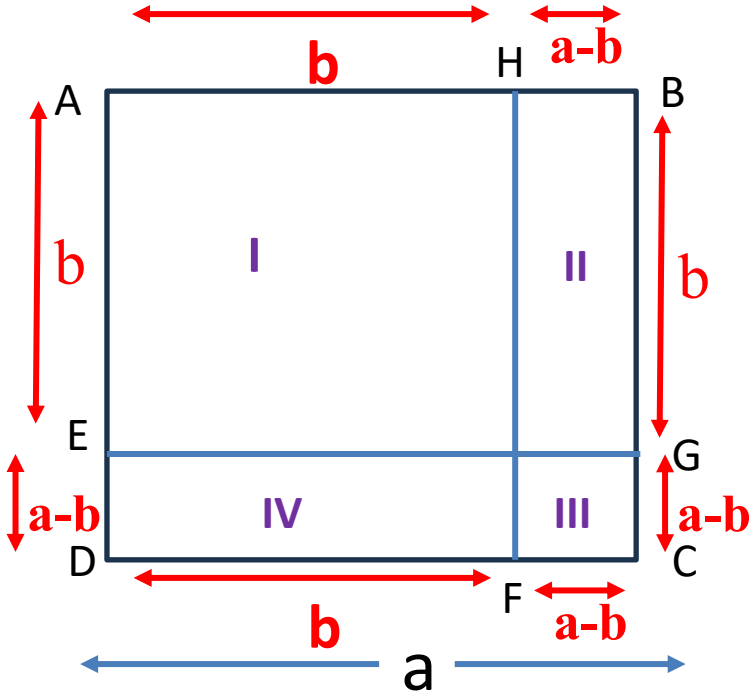
$$= a + \frac{1}{3}(\sqrt{2}a - a) = \frac{\mathbf{3a} + \sqrt{2}a - a}{\mathbf{3}}$$

$$= \frac{2a + \sqrt{2}a}{3} = \frac{a}{3}[\sqrt{2} + 2]$$

- So, the area of the square *ABCD* is= 2a.2a=4 a^2
- Now, the area of the circle is=$\pi\, \boldsymbol{r^2}$ =4.069 a^2 [$\pi = 3.141593$
  $\sqrt{2}$ = 1.414214]

Difference is very less i.e 0.069 a^2 which is negligible

Geometric proof: $(a-b)^2+2ab=(a)^2+(b)^2$



Area of square ABCD = $(\text{side})^2 = a^2$

Area of square ABCD = Area of I + Area of II + Area of III + Area of IV

$$= b^2 + b\cdot(a-b) + (a-b)^2 + b\cdot(a-b)$$

$$= b^2 + (a-b)^2 + 2b(a-b)$$

$$= b^2 + (a-b)^2 + 2ab - 2b^2$$

$$a^2 = -b^2 + (a-b)^2 + 2ab$$

$$\Rightarrow a^2 + b^2 = (a-b)^2 + 2ab$$

$$\Rightarrow (a-b)^2 + 2ab = a^2 + b^2.$$

Diagonal (hypotenuse) theorem

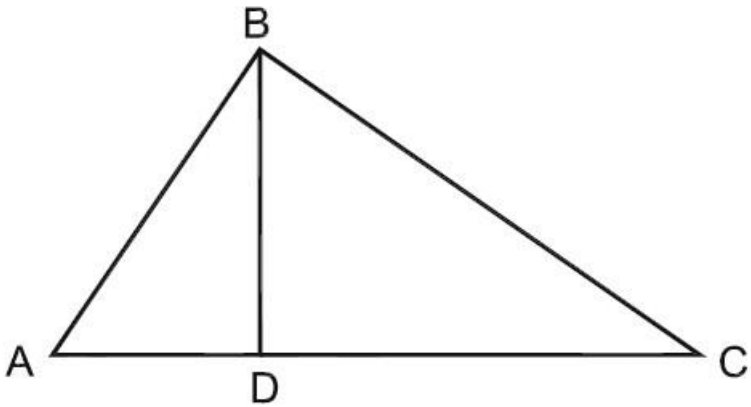


Figure 2.6: A proof based on similar triangles

- Take a right triangle **ABC**, with the **right angle at B**
- Draw **BD**, the perpendicular (altitude) from B to the hypotenuse **AC**.
- Let D be the foot of this perpendicular.

✓ Big triangle: △ABC

✓ wo smaller ones: △ADB and △BDC

❖ All three triangles are similar , Because

a) Each has a right angle.

b) Each shares an acute angle with the big triangle.

So **△ADB~△ABC,** **△BDC~△ABC**

❑ From △ADB ~ △ABC:

$$\frac{AD}{AB} = \frac{AB}{AC}$$

AB^2=AD·AC

❑ From △BDC ~ △ABC:

$$\frac{BC}{DC} = \frac{AC}{BC}$$

BC^2=DC·AC

❑ Add the two results:

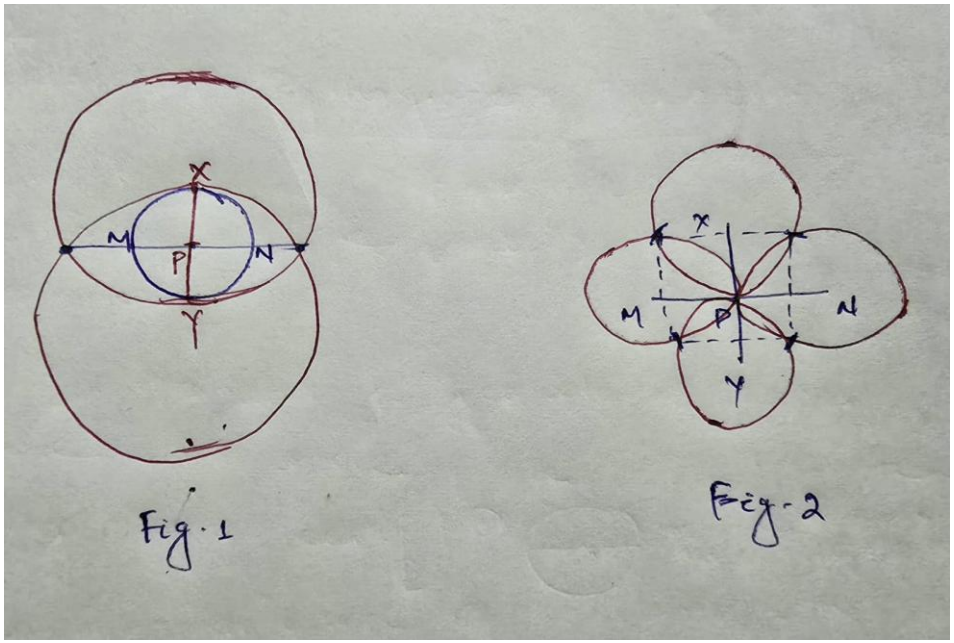
$AB^2 + BC^2$ = AD·AC + DC·AC

$AB^2 + BC^2 = AC(AD + DC)$

$AB^2 + BC^2$ =AC.AC

$\boldsymbol{AB^2 + BC^2 = AC^2}$

Baudhayana's cord-compass construction of the square



- Let us take a line segment XY & XY is of length "a" unit.
  Let "P" be the mid point of XY.
- Now keeping "p" as center of XY, let us draw a circle which will pass through the points X & Y. i.e: radius is XP.
- Now keeping X & Y as the center, we will draw two more circles whose radii are XY.

They will be cutting each other at 2 points. If we will join these two points, it is the common cord for these 2 circles.

- That will touch the inner circles in 2 points M & N.

➢ Now taking these 4 points, M,N,X,Y and radius=XP, we have to draw 4 circles.

➢ Keeping X as the center, and XP as radius , draw circle. Similarly keeping N as center & NP as radius, draw another circle. Likewise, we have to draw circles.

➢ These 4 circles will be cutting each other at different points. If you join all 4 points, that is the required square.

Rectangle into Square

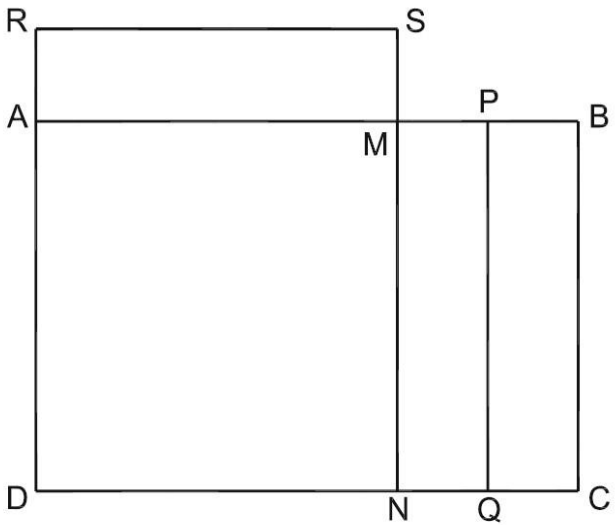


Figure 2.8: Rectangle into square

Given:
ABCD is a rectangle.
Length **AB = a**, breadth **BC = b**, with **a > b**.

- On side **AB**, mark point **M** such that:

$$AM = b$$

- On side **DC**, mark point **N** directly below **M**. Then Join **M to N**

- **AMND** forms a **square of side b**.
- The remaining right-hand portion **MBCN** is a rectangle of: Height = b & Length = a − b.
- Mark **P** as the midpoint of **MB** & Mark **Q** as the midpoint of **NC**.

$$MP = PB = \frac{a-b}{2}$$

$$\& \quad NQ = QC = \frac{a-b}{2}$$

- Cut out rectangle **PBCQ**. Its dimensions are: Height = b, $width = \frac{a-b}{2}$

So its area is $\quad b.\left(\frac{a-b}{2}\right)$

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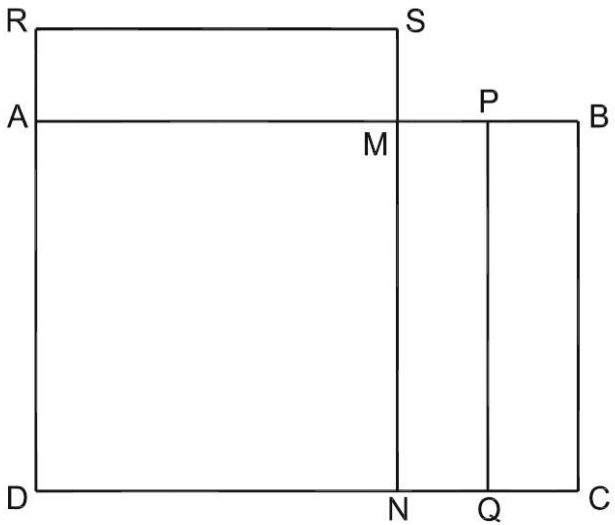

Figure 2.8: Rectangle into square

- Move this cut rectangle **PBCQ** to the top left position **ARSM**.
- Now all pieces fit together to form the new shape: $DRSMPQ(D)$
- Because we only **cut and rearranged**, the new figure has **exactly the same area** as rectangle ABCD. That area is: ab
- The new shape is **not a perfect square**, but it equals: 'difference' of the square on DQ (of side $b+(a-b)/2 = (a+b)/2$) and the square on MP (of side $(a-b)/2$)
- (Big square) − (Small square)

- Big square side: $b + \frac{a-b}{2} = \frac{a+b}{2}$
- Small square side: $\frac{a-b}{2}$
- So the area becomes: $(\frac{a+b}{2})^2$ - $(\frac{a-b}{2})^2$
- Multiply both sides by 4: $\boldsymbol{(a+b)^2-(a-b)^2=4ab}$

Short Answer Questions

- What does "Śulba" mean?
- State the Theorem of the Diagonal. Proof $\boldsymbol{AB^2 + BC^2 = AC^2}$

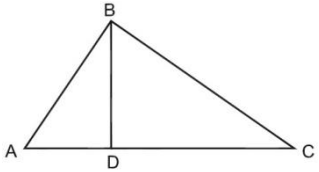

Figure 2.6: A proof based on similar triangles

- Why was this theorem important in Vedic ritual?
- Geometrically proof $(a-b)^2+2ab=(a)^2+(b)^2$
- How a circle can be constructed from square according to Baudhayana.
- Explain cord-compass construction of the square
- How a square can be constructed from a Rectangle
- The sides of a triangle are 5,12,13 units. Check if it has a right angle or not?
- The hypotenuse of a right-angled triangle is 16 units and one of the sides of triangle is 8 units. Find the measure of the third side using the Pythagoras theorem formula

Why was this theorem important in Vedic ritual?

Answer:

The Theorem of the Diagonal appears in the Śulbasūtras, ancient Indian texts concerned with the geometric construction of Vedic ritual altars. The word 'Śulba' means rope, indicating that geometry was performed using cords for measurement and construction. This theorem is historically important because it represents the earliest known statement of what is now called the Pythagoras Theorem.